

Surjya Narayan Sahoo

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Big Data Architect | AI & Data Science Specialist | Java/J2EE Expert

Results-driven professional with **19.5+ years** of experience in **Big Data, AI, Machine Learning, and Java/J2EE**, including **2 years** of specialization in AI and NLP. Proficient in building scalable, high-performance data solutions using Hadoop, Spark, cloud platforms (GCP/AWS/Azure), and cutting-edge AI models such as GPT and Hugging Face Transformers. Adept at implementing **predictive analytics, data pipelines, and real-time streaming systems** to drive actionable insights and improve business decision-making.

- Expert in designing and implementing AI-driven solutions, utilizing advanced algorithms for predictive analytics, machine learning, and real-time decision systems.
- Extensive hands-on experience with Spark MLlib for building predictive models and implementing machine learning pipelines for large-scale datasets.
- Designed and optimized scalable distributed data solutions using Datastax Cassandra to support AI-based applications.
- Migrated MapReduce workflows to Spark transformations, enhancing data processing speed to support AI model training and real-time analytics.
- Built and deployed machine learning models on Azure Databricks for automated data processing and AI-driven insights generation.
- Implemented data ingestion pipelines using Kafka Producers and Consumers, enabling real-time data flow for AI/ML applications.
- Developed real-time pre-processing pipelines using Apache Storm and Kafka to prepare data for AI-based analytics.
- Expert in NoSQL databases (Cassandra, MongoDB, HBase) to design optimized data models for AI and analytics solutions.
- Designed Spark SQL transformations for complex structured data analysis to facilitate AI model training.
- Developed and optimized machine learning models for pattern recognition, forecasting, and anomaly detection, enhancing business insights.
- Implemented AI/ML models on big data platforms, leveraging Hadoop and Spark for distributed training and predictions.
- Built and managed HBase clusters for structured storage and fast retrieval of AI-processed datasets.
- Hands-on experience developing and deploying AI models for predictive maintenance, recommendation systems, and data-driven automation.
- Utilized Kafka and Storm for real-time ingestion and pre-processing, enabling efficient data pipelines for machine learning workflows.
- Extensive experience building machine learning APIs using RESTful web services integrated with J2EE frameworks (Spring, Hibernate).
- Leveraged analytical tools to mine large datasets, uncover hidden patterns, and implement predictive analytics algorithms for actionable insights.
- Implemented custom data models and optimized queries for MongoDB and Cassandra to improve data availability for AI applications.
- Designed and implemented Spark actions for iterative algorithms and machine learning transformations to derive business-critical insights.
- Built unified platforms to integrate data from multiple sources, facilitating AI and real-time analytics.
- Extensive experience in developing machine learning pipelines using Azure ML, Databricks, and other big data platforms.

- Wrote and optimized machine learning scripts in Spark MLlib, improving execution efficiency for real-time AI decisioning.
 - Implemented scalable AI systems leveraging cloud platforms like Azure, enabling seamless deployment and monitoring.
 - Extensive experience on DBT(Data Build Tool) for data modeling
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Core Competencies

Data Engineering

- **Hadoop Ecosystem:** HDFS, Hive, Pig, Sqoop, Flume, HBase, Oozie, Spark, Kafka, YARN.
- **Streaming Analytics:** Kafka (Confluent), Spark Streaming, Storm.
- **ETL Pipelines:** Large-scale **data ingestion, transformation, and real-time data processing.**
- **Data Modeling:** DBT for building Dimensional and Fact Schema
- **Data Lakes:** Designed and migrated **200TB+ datasets** to cloud-based data lakes on **GCP, AWS, and Azure.**
- **Cluster Management:** Hands-on experience in cluster setup, performance optimization, and fault tolerance.

AI, Data Science, and Machine Learning

- **Generative AI & LLMs:** Fine-tuning **GPT-based models**, Hugging Face Transformers, and NLP pipelines.
- **Predictive Analytics:** Leveraging **Scikit-Learn**, Spark MLlib for customer insights, anomaly detection, and fraud prediction.
- **Data Science Tools:** Python, R, TensorFlow, Spark MLlib, PyTorch.
- **NLP:** Sentiment analysis, chatbots, and summarization using **GPT models** and **Hugging Face Transformers.**
- **AI Platforms:** Azure OpenAI, Google Gemini, and Form Recognizer.

Cloud Technologies

- **Google Cloud Platform (GCP):** DataProc, BigQuery, DataFusion, Cloud Storage.
- **AWS:** EC2, S3, EMR, Lambda, Redshift, Glue.
- **Azure:** Azure Data Lake, Azure Cognitive Services.

Programming & Development

- **Languages:** Scala, Python, Java, J2EE (Spring, Hibernate), C++.
 - **Frameworks:** Spark, PySpark, Kafka, Storm.
 - **Orchestration Tools:** Apache Airflow, Oozie.
 - **Databases:** Oracle, MySQL, DB2
 - **NoSQL :** Mongo DB, Cassandra, HBase
 - **Cloud DB :** - Snowflake
 - **Data Modeling:** Erwin, LucidChart, Microsoft Visio, Toad Data Modeler
 - **Reporting:** Qlikview, PowerBI, Tableau, Looker
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Key Skills

- **Big Data & Hadoop Ecosystem:** HDFS, Hive, Sqoop, Spark, Kafka, Storm, YARN, HBase, Cassandra.
- **AI & Machine Learning:** GPT Models, LLMs, Scikit-Learn, Hugging Face Transformers, NLP.

- **Cloud Platforms:** GCP (DataProc, BigQuery), AWS (EMR, Glue), Azure (AI Services, Cognitive Search).
 - **Data Pipelines, Data Modeling & ETL:** End-to-end data ingestion, transformation, and analytics workflows using Spark, Python, Scala, DBT, Erwin.
 - **Streaming & Real-Time Analytics:** Kafka brokers, Spark Streaming, Storm, real-time dashboards.
 - **Performance Optimization:** Spark and Hive tuning, distributed cluster management, CI/CD pipelines.
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SELECTED ACHIEVEMENTS

- **Predictive Analytics & AI Solutions:** Designed AI-driven predictive models using **Scikit-Learn** and **Spark MLlib**, implemented **GPT-based models** for insights in customer support and recommendation systems.
 - **NLP & LLM Integration:** Integrated **ChatGPT** and **Hugging Face Transformers** to automate responses, streamline customer service, and improve natural language processing capabilities in applications.
 - **Data Lake Architectures:** Led data migration from legacy systems and optimized big data storage on cloud platforms.
 - **Kafka and Storm Integration:** Developed Kafka brokers with custom partitions, implementing real-time streaming and pre-processing pipelines.
 - **Java/J2EE Applications:** Built web applications with Spring and Hibernate; extensive experience with WebSphere Portal.
 - **Dashboards and Visualization:** Created Node.js-based dashboards for data visualization, supporting real-time insights for FMCG clients.
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RECENT PROJECT EXPERIENCE

1. Telus Foundation Project – Wipro Technologies, Bangalore

- **Client:** Telus Communication, Canada
 - **Role:** Designed and implemented advanced AI and data processing solutions, integrating LLMs, GPT-based models, Hugging Face libraries, and Google Gemini to drive comprehensive data analytics and enhanced customer interaction handling.
 - **Technologies:** GCP, DataFusion, DataProc, BigQuery, GCP Data Lake, Google Gemini, Hugging Face, GPT models, LLMs
 - **Implementation Details:**
 1. **LLM and GPT Deployment:** Integrated LLMs and GPT models within Google Gemini to automate customer support, conduct sentiment analysis, and predict network issues, significantly improving response accuracy and operational efficiency.
 2. **Hugging Face Integration:** Utilized Hugging Face's transformer models for fine-tuning NLP applications, enabling advanced customer feedback analysis and predictive maintenance capabilities.
 3. **Data Lake Architecture:** Designed and deployed a scalable GCP-based Data Lake to manage vast customer and operational datasets, ensuring seamless AI model integration for real-time insights and enhanced decision-making.
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2. Financial Crime Data Mart (FCDM)

- **Client:** US Bank, USA
 - **Role:** Designed and developed data ingestion and processing pipelines to consolidate fraud insights and risk management.
 - **Technologies:** Spark, Databricks, Kafka, Python, Tableau
 - **Key Contributions:**
 - Developed scalable data pipelines for fraud detection and compliance analytics.
 - Optimized ETL processes for real-time transaction monitoring and anomaly detection.
 - Improved data governance and reporting mechanisms to enhance regulatory compliance.
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3. Uplight Connect Platform

- **Client:** Uplight, USA (Google End Client)
 - **Role:** Architected and optimized data ingestion and transformation pipelines on GCP to support energy management and analytics.
 - **Technologies:** GCP, DataFusion, DataProc, BigQuery
 - **Key Contributions:**
 - Designed scalable data pipelines to process real-time energy usage data.
 - Enabled seamless integration with third-party analytics tools for enhanced energy efficiency insights.
 - Implemented cost-optimized cloud storage solutions to improve performance and reduce processing costs.
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4. RBE (Rule-Based Engine) – BNY Mellon

- **Client:** BNY Mellon, USA
 - **Role:** Designed and developed data ingestion pipelines and Data Lake architecture, implementing a billing management system with Spark for data transformation and automated scheduling.
 - **Technologies:** Spark, Airflow, AWS, Hive, Erwin
 - **Key Contributions:**
 - Implemented an efficient rule-based billing system for financial data processing.
 - Developed scalable ETL pipelines for real-time and batch data processing.
 - Automated workflows using Apache Airflow to optimize billing cycles and reduce manual intervention.
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5. Nike Data Foundation

- **Client:** Nike, USA
 - **Role:** Created and optimized the data lake architecture with Airflow-managed pipelines to support advanced analytics and reporting.
 - **Technologies:** AWS, Spark, Airflow, Hive, Snowflake, Toad Data Modeler
 - **Key Contributions:**
 - Designed an AWS-based Data Lake to store and process large-scale customer and sales data.
 - Developed automated data pipelines with Apache Airflow, improving data availability and reliability.
 - Enabled Snowflake-based data warehousing solutions for advanced analytics and reporting.
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ADDITIONAL PROJECTS

- **EISL (UBS, USA):** Kafka-based data orchestration and Spark-based data aggregation.
- **Chevron EDAP:** Built enterprise analytics platforms for data filtration and business dashboards.
- **COOP NDX Loyalty:** Migrated and transformed loyalty data using Spark SQL.
- **AT&T RAPTOR:** Deployed Hadoop/Spark/MongoDB-based service assurance systems.
- **Vodafone Socio Data Archival:** Architected CRM archival and analytics systems using Hadoop.

CORE SKILLS

- **Big Data Technologies:** Hadoop, Spark, Kafka, Hive, MongoDB, Cassandra
- **AI & ML Tools:** GPT, LLMs, ChatGPT, Scikit-Learn, Spark MLlib, Hugging Face Transformers
- **Cloud Platforms:** GCP, AWS, Azure
- **Programming Languages:** Scala, Python, Java, Spark SQL
- **Performance Optimization:** Hive tuning, Spark optimization, disaster recovery, CI/CD pipelines

EDUCATION

- **Executive Programme in Business Management (2011)** - IIM Calcutta (Grade: "Good")
- **Master of Computer Applications (MCA)(2000-2003)** - Utkal University

PROFESSIONAL EXPERIENCE

- **Wipro Technologies, Bangalore** – Senior Architect(Oct,2018-Nov,2024)
- **Nielsen India Pvt. Ltd., Bangalore** – Associate Director(July,2017-Feb,2018)
- **TechMahindra Ltd., Bangalore** – Solution Architect(April,2016-April,2017)
- **Capgemini India Pvt Ltd., Bangalore** – Manager(Nov,2014 –Nov,2015)
- **Accion Labs India Pvt Ltd., Bangalore** – Senior Manager(Dec,2012 – Oct,2014)
- **ITC Infotech India Ltd., Bangalore** – Lead Consultant(Aug,2011 – Sept,2012)
- **Symphony Services, Bangalore** – Tech Lead(Aug,2010 – Aug,2011)
- **MphasiS, Bangalore** – Module Lead(Sept,2009 – July,2010)
- **Exilant Consulting Pvt. Ltd., Bangalore** – SSE(March,2009 – July,2009)
- **Fujitsu Consulting India Pvt. Ltd., Noida** – SSE(Jan,2008 – Jan,2009)
- **L&T Infotech Pvt. Ltd., Mumbai** – Software Engineer(July,2007 – Jan,2008)
- **Patni Computers Pvt. Ltd., Mumbai** – Software Engineer(Nov,2005 – July,2007)